

1.Requirement Engineering

Step No.	Process Step (PDCA)	Input	Tools / Methods Used	Critical Parameters	Output / Check
1	PLAN – Identify Problem / Objective	Bottling plant issues, defect list, bottle samples, process flow	Requirement analysis, user interviews, observation, document study	Defect types (cracks, impurities, fill-level, cap issues), production speed, inspection accuracy	Clear problem statement & defined project objective
2	PLAN – Analyse Root Causes & Develop Plan	Problem statement, sample defective bottles, existing system data	Root cause analysis (Fishbone, 5 Why), process mapping, feasibility analysis	Major defect sources, lighting issues, camera angles, conveyor speed, false rejects	Implementation plan, inspection strategy & system design
3	DO – Implement Solution (Pilot Implementation)	Hardware (camera, lights), software (OpenCV), test bottles, conveyor section	Pilot setup, prototype build, algorithm testing, calibration	Camera resolution, lighting uniformity, ROI selection, processing speed	Working pilot model detecting main defects
4	CHECK – Evaluate Performance	Pilot test results, defect detection data, accuracy logs	Performance testing, confusion matrix, accuracy measurement, stress test	Detection accuracy > 95%, false reject < 2%, FPS > 25	Verified performance & gaps identified
5	ACT – Standardize or Improve Further	Test feedback, error logs, operator suggestions	Refinement, parameter tuning, additional training, optimization	Improved algorithm stability, better ROI tuning, noise reduction	Standardized inspection process ready for deployment
6	ACT – Full Rollout & Monitoring	Final stable system, PLC/rejector integration, SOP	Full deployment, continuous monitoring, maintenance plan	Long-term stability, system uptime, maintenance cycle	Fully implemented machine vision inspection system with continuous monitoring

2. Concept Design

Step	Process Step	Input	Tools / Methods	Critical Parameters	Output
1	PLAN – Identify Problem / Objective	Defect list, sample bottles	Brainstorming, requirement study	Type of defects, accuracy target	Clear inspection objective
2	PLAN – Analyse Root Causes & Develop Plan	Process study, manual inspection issues	5-Why, Fishbone, layout study	Lighting issues, line speed, bottle variation	Project plan + concept design
3	DO – Implement Pilot Solution	Camera setup, lighting, sample bottles	Mounting trial, OpenCV prototype	Camera angle, exposure, preprocessing	Working pilot demo
4	CHECK – Evaluate Performance	Pilot results, defect logs	Accuracy test, tuning	Accuracy %, false rejects	Performance report
5	ACT – Standardize / Improve	Optimized settings	SOP creation, code improvement	Stable accuracy, robustness	Final optimized design
6	ACT – Full Rollout & Monitoring	Standardized system	Final install, PLC link, dashboard	Uptime, reject rate	Full production system + monitoring

3.Embodiment design

Step	Process Step	Input	Tools / Methods	Critical Parameters	Output
1	PLAN – Identify Problem / Objective	Defect types, sample bottles, plant requirements	Requirement study, bottle analysis	What defects to detect, speed, accuracy target	Clear embodiment design goal
2	PLAN – Analyse Root Causes & Develop Plan	Process issues, lighting variation, bottle movement	Fishbone, 5-Why, layout study	Camera position, vibration, lighting uniformity	Embodiment concept (mechanical + vision layout)
3	DO – Implement Pilot Solution	Camera, lens, lighting, frames, mounting parts	CAD modelling, prototype build, OpenCV testing	Mount stability, angle, exposure, detection logic	Working embodiment prototype
4	CHECK – Evaluate Performance	Pilot test data, images, accuracy logs	Test validation, measurement, tuning	Accuracy %, repeatability, cycle time	Performance evaluation report
5	ACT – Standardize / Improve	Improved design, optimized settings	Redesign parts, refine software, SOP creation	Durability, alignment, stable detection	Final embodiment design ready
6	ACT – Full Rollout & Monitoring	Approved design, production needs	Full installation, PLC integration, monitoring	Uptime, reject rate, maintenance frequency	Complete working system with continuous monitoring

4.Detail design

Step	Process Step	Input	Tools / Methods	Critical Parameters	Output
1	PLAN – Identify Problem / Objective	Required detection accuracy, plant speed, bottle defects	Requirement study, specification review	Accuracy, cycle time, resolution, lighting needs	Clear detailed design target
2	PLAN – Analyse Root Causes & Develop Plan	Mechanical constraints, lighting issues, bottle variations	Tolerance analysis, error analysis, risk assessment	Camera FOV, lens selection, mounting height, light intensity	Detailed design plan (mechanical + software)
3	DO – Implement Pilot Detail Design	CAD models, wiring diagrams, software flow	3D modelling, circuit design, OpenCV algorithm coding	Part dimensions, sensor placement, exposure settings	Detailed prototype ready for testing
4	CHECK – Evaluate Performance	Test data, prototype results	Tolerance check, accuracy test, defect validation	Frame rate, processing time, false reject rate	Detailed design verification report
5	ACT – Standardize / Improve Further	Updated design files, optimized code	Design refinement, documentation, SOP creation	Repeatability, robustness, alignment consistency	Final detailed design (approved)
6	ACT – Full Rollout & Monitoring	Final approved design	Production installation, PLC integration, QC monitoring	System uptime, reject accuracy, maintenance intervals	Full working system with continuous monitoring

5.Sourcing

Step	Process Step	Input	Tools / Methods	Critical Parameters	Output
1	PLAN – Identify Problem / Objective	Required camera, lens, lighting, PC specs, budget	Requirement study, vendor list	Quality, cost, lead time	Clear sourcing requirements
2	PLAN – Analyse Root Causes & Develop Plan	Supplier performance data, technical specifications	Vendor comparison, risk analysis	Warranty, support, delivery reliability	Sourcing strategy & selected suppliers
3	DO – Implement Solution (Pilot Purchasing)	Sample units, quotes, test components	Trial purchase, testing components on prototype	Compatibility, build quality, stability	Pilot hardware sourced and tested
4	CHECK – Evaluate Performance	Component test results, inspection logs	Performance check, defect test, reliability test	Accuracy, heat tolerance, lifetime, signal quality	Evaluation report (pass/fail)
5	ACT – Standardize / Improve Further	Best-performing components, pricing	Negotiation, standard part list creation	Consistent availability, optimized cost	Standardized sourcing list (BOM)
6	ACT – Full Rollout & Monitoring	Final BOM, approved vendors	Bulk ordering, inventory tracking	Stock levels, delivery time, quality consistency	Complete sourcing system with continuous monitoring

6.Prototype Building

Step	Process Step	Input	Tools / Methods	Critical Parameters	Output
1	PLAN – Identify Problem / Objective	Prototype requirements, defect types, bottle samples	Requirement analysis, concept sketching	Accuracy, speed, inspection zones	Clear prototype objective
2	PLAN – Analyse Root Causes & Develop Plan	Layout constraints, hardware needs, lighting issues	Feasibility study, root-cause analysis	Camera angle, light uniformity, stability	Prototype build plan
3	DO – Implement Solution (Pilot Prototype)	Camera, lens, lighting, frame, wiring	Mechanical assembly, coding (OpenCV), mounting trials	Positioning, exposure, vibration control	Working pilot prototype
4	CHECK – Evaluate Performance	Test images, detection accuracy data	Performance testing, tuning, defect validation	Accuracy %, false rejects, frame rate	Prototype evaluation report
5	ACT – Standardize / Improve Further	Improvement points, updated design	Redesign frame, adjust code, refine lighting	Robustness, repeatability, alignment	Improved & stable prototype
6	ACT – Full Rollout & Monitoring	Final prototype, approved performance	Integration to line, SOP creation, monitoring plan	Uptime, reliability, detection consistency	Prototype ready for production rollout

7. Prototype testing

Step	Process Step	Input	Tools / Methods	Critical Parameters	Output
1	PLAN – Identify Problem / Objective	Test requirements, defect list, sample bottles	Requirement review, test planning	Accuracy target, cycle time, defect coverage	Clear prototype testing objective
2	PLAN – Analyse Root Causes & Develop Plan	Prototype issues, earlier test failures	Root-cause analysis, risk study	Lighting variation, motion blur, bottle rotation	Test plan with conditions & parameters
3	DO – Implement Solution (Pilot Testing)	Prototype setup, test dataset, trial samples	Test execution, logging, image capture	Exposure, threshold settings, camera stability	Raw test results collected
4	CHECK – Evaluate Performance	Accuracy data, false positives/negatives	Confusion matrix, performance analysis	Accuracy %, repeatability, processing time	Testing evaluation report
5	ACT – Standardize / Improve Further	Identified issues, tuning inputs	Parameter optimization, code refinement	Reduced false alarms, stable detection	Improved prototype ready for final test
6	ACT – Full Rollout & Monitoring	Final validated prototype	Full-line trial, operator feedback, monitoring	Long-run stability, reject rate, uptime	Prototype approved for production rollout

8.Iteration for optimization

Step	Process Step	Input	Tools / Methods	Critical Parameters	Output
1	PLAN – Identify Problem / Objective	Current system performance, defects missed, false rejects	Performance review, log analysis	Accuracy gap, speed gap, stability issues	Clear optimization goal
2	PLAN – Analyse Root Causes & Develop Plan	Failure cases, images, lighting changes	Root-cause analysis, algorithm comparison	Lighting hotspots, blur, threshold mismatch	Optimization plan & parameter list
3	DO – Implement Solution (Pilot Optimization)	Updated parameters, modified code, improved setup	Tuning, re-coding, lighting adjustment	Exposure, filter strength, ROI accuracy	Optimized pilot version
4	CHECK – Evaluate Performance	New test results, accuracy data	A/B comparison, validation testing	Improvement % vs old system	Optimization evaluation report
5	ACT – Standardize / Improve Further	Best-performing settings	Documenting optimized parameters, update SOP	Consistency, repeatability	Standardized optimized settings
6	ACT – Full Rollout & Monitoring	Approved optimized version	Line integration, continuous monitoring	Long-term stability, reject trends	Fully optimized system in production

9.Documenting for sharing

Step	Process Step	Input	Tools / Methods	Critical Parameters	Output
1	PLAN – Identify Problem / Objective	Project details, test results, design data	Requirement review, content planning	Accuracy of information, clarity	Objective of document clearly defined
2	PLAN – Analyse Root Causes & Develop Plan	Missing data, unclear sections, feedback	Gap analysis, review meetings	Completeness, correctness, readability	Structured document outline
3	DO – Implement Solution (Pilot Document)	Draft content, tables, images	Document creation, formatting	Consistency, correct terminology	First draft of sharable document
4	CHECK – Evaluate Performance	Draft document, review comments	Peer review, proofreading	Errors, formatting issues, clarity	Improved and corrected document
5	ACT – Standardize / Improve Further	Final draft, updated info	Version control, standard templates	Alignment with standards, accuracy	Final polished document
6	ACT – Full Rollout & Monitoring	Approved final document	Distribution, storage, periodic updates	Accessibility, version tracking	Official shared document for stakeholders